

# Assignment 7: GLMs (Linear Regressios, ANOVA, & t-tests)

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## OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on generalized linear models.

## Directions

1. Rename this file `<FirstLast>_A07_GLMs.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change “Student Name” on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Be sure to **answer the questions** in this assignment document.
5. When you have completed the assignment, **Knit** the text and code into a single PDF file.

**!! AI generated code should NOT be used in this assignment**

## Set up your session

1. Set up your session. Check your working directory. Load the tidyverse, agricolae and other needed packages. Import the *raw* NTL-LTER raw data file for chemistry/physics (NTL-LTER\_Lake\_ChemistryPhysics\_Raw.csv). Set date columns to date objects.
2. Build a ggplot theme and set it as your default theme.

*#1*

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
```

```
## v dplyr      1.1.4      v readr      2.1.5
```

```
## v forcats    1.0.0      v stringr   1.5.1
```

```
## v ggplot2    3.5.1      v tibble    3.2.1
```

```
## v lubridate  1.9.3      v tidyr     1.3.1
```

```
## v purrr      1.0.2
```

```
## -- Conflicts ----- tidyverse_conflicts() --
```

```
## x dplyr::filter() masks stats::filter()
```

```
## x dplyr::lag()     masks stats::lag()
```

```
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(agricolae)
```

```
library(lubridate)
```

```
options(scipen = 4)
```

```
library(here)
```

```
## here() starts at /home/guest/EDE_Fall2025
```

```
here()
```

```
## [1] "/home/guest/EDE_Fall2025"
```

```
chem.phys <- read.csv(here("Data/Raw/NTL-LTER_Lake_ChemistryPhysics_Raw.csv"), stringsAsFactors = TRUE)
chem.phys$sampdate <- as.Date(chem.phys$sampdate, format = "%m/%d/%y")
class(chem.phys$sampdate)
```

```
## [1] "Date"
```

```
#2
mytheme <- theme_classic(base_size = 13) +
  theme(axis.text = element_text(color = "black"),
        plot.background = element_rect(fill = "white", color = NA),
        legend.position = "right")
theme_set(mytheme)
```

## Simple regression

Our first research question is: Does mean lake temperature recorded during July change with depth across all lakes?

3. State the null and alternative hypotheses for this question: > Answer: H0: Mean lake temperature recorded during July does not change with depth across all lakes Ha: Mean lake temperature recorded during July changes with depth across all lakes
4. Wrangle your NTL-LTER dataset with a pipe function so that the records meet the following criteria:
  - Only dates in July.
  - Only the columns: lakename, year4, daynum, depth, temperature\_C
  - Only complete cases (i.e., remove NAs)
5. Visualize the relationship among the two continuous variables with a scatter plot of temperature by depth. Add a smoothed line showing the linear model, and limit temperature values from 0 to 35 °C. Make this plot look pretty and easy to read.

```
#4
chem.phys.month <- mutate(chem.phys, month = month(sampdate))

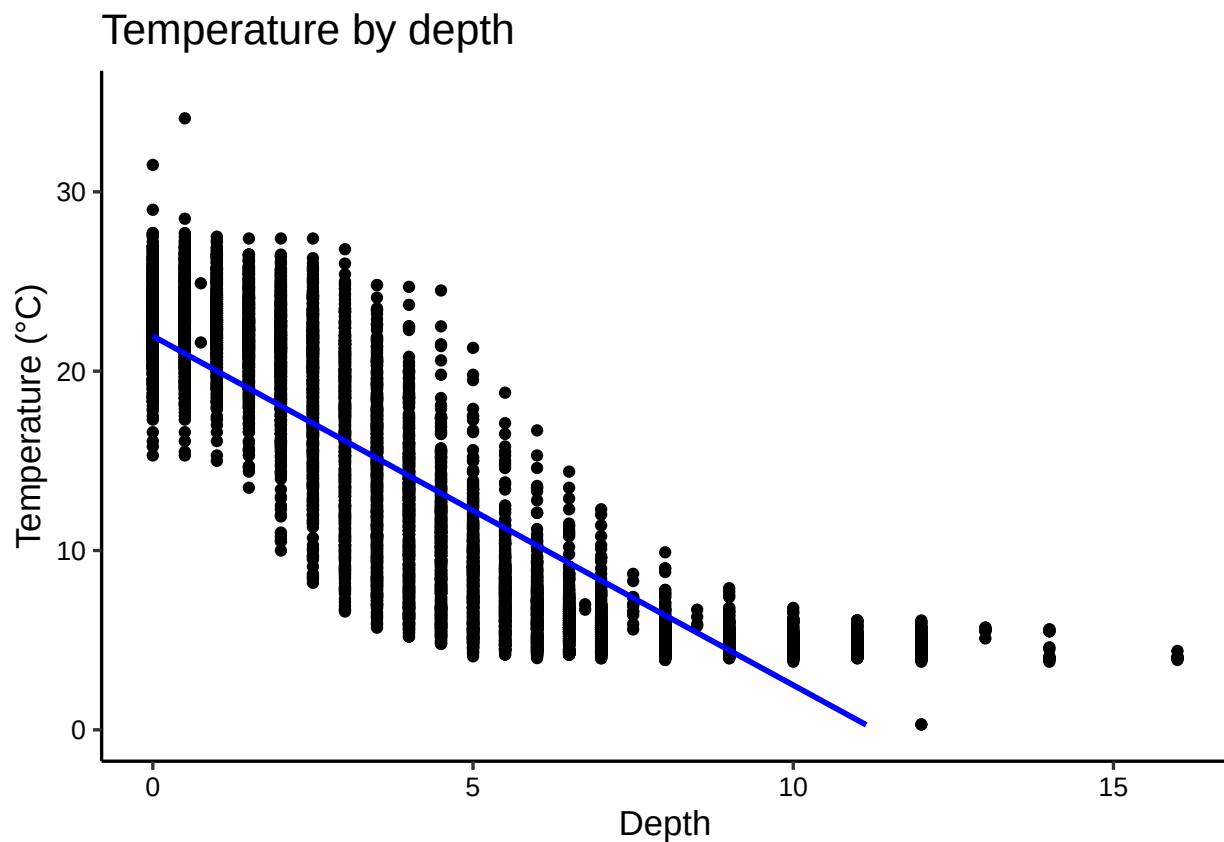
chem.phys.july <- chem.phys.month %>%
  filter(month == 7) %>%
  select(lakename, year4, daynum, depth, temperature_C) %>%
  na.omit()
```

```
#5
temp.depth <- ggplot(chem.phys.july, aes(x = depth, y = temperature_C)) +
  geom_point() +
  geom_smooth(method = "lm", se = FALSE, color = "blue") +
```

```
ylim(0, 35)+
labs(title = "Temperature by depth",
     x = "Depth",
     y = "Temperature (°C)")
print(temp.depth)
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

```
## Warning: Removed 24 rows containing missing values or values outside the scale range
## ('geom_smooth()').
```



6. Interpret the figure. What does it suggest with regards to the response of temperature to depth? Do the distribution of points suggest anything about the linearity of this trend?

Answer: The general trend of the figure suggests that there is a negative relationship between temperature and depth. However, the distribution of points does not suggest a clear linearity of the trend. Stacked y value points for each x value points does not show clear linearity.

7. Perform a linear regression to test the relationship and display the results.

```
#7
temp.depth.regression <- lm(chem.phys.july$temperature_C ~
                           chem.phys.july$depth)
summary(temp.depth.regression)
```

```
##
## Call:
## lm(formula = chem.phys.july$temperature_C ~ chem.phys.july$depth)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -9.5173 -3.0192  0.0633  2.9365 13.5834
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    21.95597    0.06792   323.3  <2e-16 ***
## chem.phys.july$depth -1.94621    0.01174  -165.8  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.835 on 9726 degrees of freedom
## Multiple R-squared:  0.7387, Adjusted R-squared:  0.7387
## F-statistic: 2.75e+04 on 1 and 9726 DF, p-value: < 2.2e-16
```

8. Interpret your model results in words. Include how much of the variability in temperature is explained by changes in depth, the degrees of freedom on which this finding is based, and the statistical significance of the result. Also mention how much temperature is predicted to change for every 1m change in depth.

Answer: The results show that about 74% of the variability in temperature is explained by changes in depth. There are 9726 degrees of freedom of which this finding is based. The p-value (less than 0.05) indicates that the relationship between temperature and depth is statistically significant. Results show that for every 1 meter change in depth, the temperature will change by 1.95 degrees Celsius.

---

## Multiple regression

Let's tackle a similar question from a different approach. Here, we want to explore what might the best set of predictors for lake temperature in July across the monitoring period at the North Temperate Lakes LTER.

9. Run an AIC to determine what set of explanatory variables (year4, daynum, depth) is best suited to predict temperature.
10. Run a multiple regression on the recommended set of variables.

```
#9
temp.aic <- lm(data = chem.phys.july, temperature_C ~ depth + year4 + daynum)
step(temp.aic)
```

```
## Start: AIC=26065.53
## temperature_C ~ depth + year4 + daynum
##
##           Df Sum of Sq    RSS   AIC
## <none>                141687 26066
```

```
## - year4    1      101 141788 26070
## - daynum   1      1237 142924 26148
## - depth    1     404475 546161 39189
```

```
##
## Call:
## lm(formula = temperature_C ~ depth + year4 + daynum, data = chem.phys.july)
##
## Coefficients:
## (Intercept)      depth      year4      daynum
##    -8.57556    -1.94644     0.01134     0.03978
```

```
#10
temp.model <- lm(data = chem.phys.july, temperature_C ~ depth + year4 + daynum)
summary(temp.model)
```

```
##
## Call:
## lm(formula = temperature_C ~ depth + year4 + daynum, data = chem.phys.july)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -9.6536 -3.0000  0.0902  2.9658 13.6123
##
## Coefficients:
##              Estimate Std. Error  t value Pr(>|t|)
## (Intercept) -8.575564   8.630715  -0.994  0.32044
## depth       -1.946437   0.011683 -166.611 < 2e-16 ***
## year4        0.011345   0.004299   2.639  0.00833 **
## daynum       0.039780   0.004317   9.215 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.817 on 9724 degrees of freedom
## Multiple R-squared:  0.7412, Adjusted R-squared:  0.7411
## F-statistic: 9283 on 3 and 9724 DF, p-value: < 2.2e-16
```

11. What is the final set of explanatory variables that the AIC method suggests we use to predict temperature in our multiple regression? How much of the observed variance does this model explain? Is this an improvement over the model using only depth as the explanatory variable?

Answer: The results show that, the final set of explanatory variables include all three variables (daynum, depth, and year4) as that combination (not removing any variables) already gave us the lowest AIC.

---

## Analysis of Variance

12. Now we want to see whether the different lakes have, on average, different temperatures in the month of July. Run an ANOVA test to complete this analysis. (No need to test assumptions of normality or similar variances.) Create two sets of models: one expressed as an ANOVA models and another expressed as a linear model (as done in our lessons).

#12

```
temp.anova <- aov(data = chem.phys.july, temperature_C ~ lakename)
summary(temp.anova)
```

```
##              Df Sum Sq Mean Sq F value Pr(>F)
## lakename      8  21642   2705.2     50 <2e-16 ***
## Residuals   9719 525813     54.1
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
temp.anova2 <- lm(data = chem.phys.july, temperature_C ~ lakename)
summary(temp.anova2)
```

```
##
## Call:
## lm(formula = temperature_C ~ lakename, data = chem.phys.july)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -10.769  -6.614  -2.679   7.684  23.832
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      17.6664     0.6501  27.174 < 2e-16 ***
## lakenameCrampton Lake      -2.3145     0.7699  -3.006 0.002653 **
## lakenameEast Long Lake     -7.3987     0.6918 -10.695 < 2e-16 ***
## lakenameHummingbird Lake   -6.8931     0.9429  -7.311 2.87e-13 ***
## lakenamePaul Lake         -3.8522     0.6656  -5.788 7.36e-09 ***
## lakenamePeter Lake        -4.3501     0.6645  -6.547 6.17e-11 ***
## lakenameTuesday Lake     -6.5972     0.6769  -9.746 < 2e-16 ***
## lakenameWard Lake         -3.2078     0.9429  -3.402 0.000672 ***
## lakenameWest Long Lake    -6.0878     0.6895  -8.829 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 7.355 on 9719 degrees of freedom
## Multiple R-squared:  0.03953,    Adjusted R-squared:  0.03874
## F-statistic:    50 on 8 and 9719 DF,  p-value: < 2.2e-16
```

13. Is there a significant difference in mean temperature among the lakes? Report your findings.

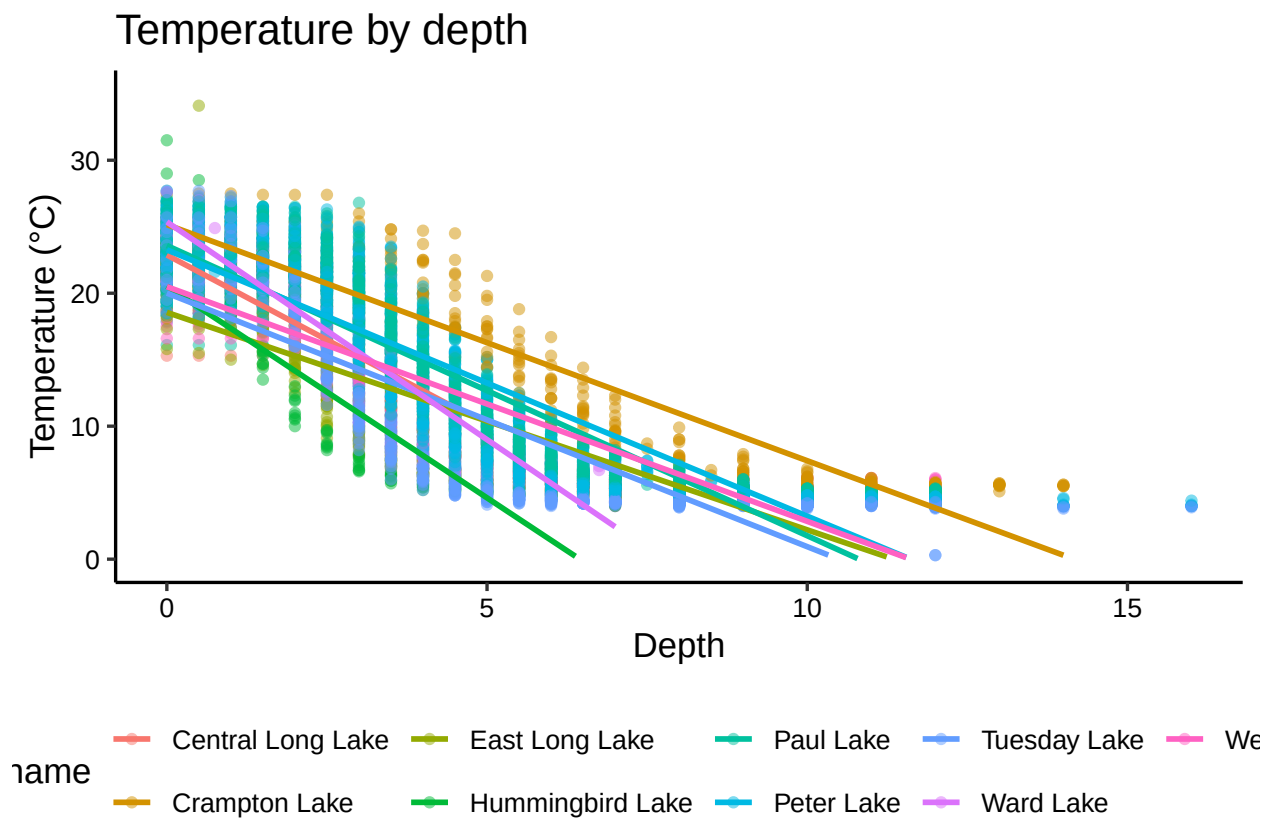
Answer: Yes, we can see from the results of the linear model that the mean of different lakes are very different. Each lake has a different mean value from each other, as can be seen from the “estimated std.” column, where the differences in mean are shown.

14. Create a graph that depicts temperature by depth, with a separate color for each lake. Add a `geom_smooth` (method = “lm”, se = FALSE) for each lake. Make your points 50 % transparent. Adjust your y axis limits to go from 0 to 35 degrees. Clean up your graph to make it pretty.

```
#14.
temp.lake <- ggplot(chem.phys.july, aes(x = depth, y = temperature_C, color = lakename)) +
  geom_point(alpha = 0.5) +
  geom_smooth(method = "lm", se = FALSE)+
  ylim(0, 35)+
  labs(title = "Temperature by depth",
       x = "Depth",
       y = "Temperature (°C)") +
  theme(
    legend.position = "bottom")
print(temp.lake)
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

```
## Warning: Removed 73 rows containing missing values or values outside the scale range
## ('geom_smooth()').
```



15. Use the Tukey's HSD test to determine which lakes have different means.

```
#15
tukey.lake.mean <- TukeyHSD(temp.anova)
tukey.lake.mean
```

```
## Tukey multiple comparisons of means
## 95% family-wise confidence level
##
## Fit: aov(formula = temperature_C ~ lakename, data = chem.phys.july)
##
## $lakename
##
```

|                                    | diff       | lwr        | upr        | p adj     |
|------------------------------------|------------|------------|------------|-----------|
| Crampton Lake-Central Long Lake    | -2.3145195 | -4.7031913 | 0.0741524  | 0.0661566 |
| East Long Lake-Central Long Lake   | -7.3987410 | -9.5449411 | -5.2525408 | 0.0000000 |
| Hummingbird Lake-Central Long Lake | -6.8931304 | -9.8184178 | -3.9678430 | 0.0000000 |
| Paul Lake-Central Long Lake        | -3.8521506 | -5.9170942 | -1.7872070 | 0.0000003 |
| Peter Lake-Central Long Lake       | -4.3501458 | -6.4115874 | -2.2887042 | 0.0000000 |
| Tuesday Lake-Central Long Lake     | -6.5971805 | -8.6971605 | -4.4972005 | 0.0000000 |
| Ward Lake-Central Long Lake        | -3.2077856 | -6.1330730 | -0.2824982 | 0.0193405 |
| West Long Lake-Central Long Lake   | -6.0877513 | -8.2268550 | -3.9486475 | 0.0000000 |
| East Long Lake-Crampton Lake       | -5.0842215 | -6.5591700 | -3.6092730 | 0.0000000 |
| Hummingbird Lake-Crampton Lake     | -4.5786109 | -7.0538088 | -2.1034131 | 0.0000004 |
| Paul Lake-Crampton Lake            | -1.5376312 | -2.8916215 | -0.1836408 | 0.0127491 |
| Peter Lake-Crampton Lake           | -2.0356263 | -3.3842699 | -0.6869828 | 0.0000999 |
| Tuesday Lake-Crampton Lake         | -4.2826611 | -5.6895065 | -2.8758157 | 0.0000000 |
| Ward Lake-Crampton Lake            | -0.8932661 | -3.3684639 | 1.5819317  | 0.9714459 |
| West Long Lake-Crampton Lake       | -3.7732318 | -5.2378351 | -2.3086285 | 0.0000000 |
| Hummingbird Lake-East Long Lake    | 0.5056106  | -1.7364925 | 2.7477137  | 0.9988050 |
| Paul Lake-East Long Lake           | 3.5465903  | 2.6900206  | 4.4031601  | 0.0000000 |
| Peter Lake-East Long Lake          | 3.0485952  | 2.2005025  | 3.8966879  | 0.0000000 |
| Tuesday Lake-East Long Lake        | 0.8015604  | -0.1363286 | 1.7394495  | 0.1657485 |
| Ward Lake-East Long Lake           | 4.1909554  | 1.9488523  | 6.4330585  | 0.0000002 |
| West Long Lake-East Long Lake      | 1.3109897  | 0.2885003  | 2.3334791  | 0.0022805 |
| Paul Lake-Hummingbird Lake         | 3.0409798  | 0.8765299  | 5.2054296  | 0.0004495 |
| Peter Lake-Hummingbird Lake        | 2.5429846  | 0.3818755  | 4.7040937  | 0.0080666 |
| Tuesday Lake-Hummingbird Lake      | 0.2959499  | -1.9019508 | 2.4938505  | 0.9999752 |
| Ward Lake-Hummingbird Lake         | 3.6853448  | 0.6889874  | 6.6817022  | 0.0043297 |
| West Long Lake-Hummingbird Lake    | 0.8053791  | -1.4299320 | 3.0406903  | 0.9717297 |
| Peter Lake-Paul Lake               | -0.4979952 | -1.1120620 | 0.1160717  | 0.2241586 |
| Tuesday Lake-Paul Lake             | -2.7450299 | -3.4781416 | -2.0119182 | 0.0000000 |
| Ward Lake-Paul Lake                | 0.6443651  | -1.5200848 | 2.8088149  | 0.9916978 |
| West Long Lake-Paul Lake           | -2.2356007 | -3.0742314 | -1.3969699 | 0.0000000 |
| Tuesday Lake-Peter Lake            | -2.2470347 | -2.9702236 | -1.5238458 | 0.0000000 |
| Ward Lake-Peter Lake               | 1.1423602  | -1.0187489 | 3.3034693  | 0.7827037 |
| West Long Lake-Peter Lake          | -1.7376055 | -2.5675759 | -0.9076350 | 0.0000000 |
| Ward Lake-Tuesday Lake             | 3.3893950  | 1.1914943  | 5.5872956  | 0.0000609 |
| West Long Lake-Tuesday Lake        | 0.5094292  | -0.4121051 | 1.4309636  | 0.7374387 |
| West Long Lake-Ward Lake           | -2.8799657 | -5.1152769 | -0.6446546 | 0.0021080 |

```
peter.similar <- HSD.test(temp.anova, "lakename", group = TRUE)
peter.similar
```

```
## $statistics
## MSerror Df Mean CV
## 54.1016 9719 12.72087 57.82135
##
## $parameters
## test name.t ntr StudentizedRange alpha
## Tukey lakename 9 4.387504 0.05
```



```
##
## $means
##           temperature_C      std      r      se Min  Max    Q25   Q50
## Central Long Lake      17.66641 4.196292  128 0.6501298 8.9 26.8 14.400 18.40
## Crampton Lake          15.35189 7.244773  318 0.4124692 5.0 27.5  7.525 16.90
## East Long Lake         10.26767 6.766804  968 0.2364108 4.2 34.1  4.975  6.50
## Hummingbird Lake       10.77328 7.017845  116 0.6829298 4.0 31.5  5.200  7.00
## Paul Lake              13.81426 7.296928 2660 0.1426147 4.7 27.7  6.500 12.40
## Peter Lake             13.31626 7.669758 2872 0.1372501 4.0 27.0  5.600 11.40
## Tuesday Lake          11.06923 7.698687 1524 0.1884137 0.3 27.7  4.400  6.80
## Ward Lake              14.45862 7.409079  116 0.6829298 5.7 27.6  7.200 12.55
## West Long Lake         11.57865 6.980789 1026 0.2296314 4.0 25.7  5.400  8.00
##           Q75
## Central Long Lake    21.000
## Crampton Lake        22.300
## East Long Lake       15.925
## Hummingbird Lake     15.625
## Paul Lake            21.400
## Peter Lake           21.500
## Tuesday Lake         19.400
## Ward Lake            23.200
## West Long Lake       18.800
##
## $comparison
## NULL
##
## $groups
##           temperature_C groups
## Central Long Lake      17.66641      a
## Crampton Lake          15.35189     ab
## Ward Lake              14.45862     bc
## Paul Lake              13.81426      c
## Peter Lake             13.31626      c
## West Long Lake         11.57865      d
## Tuesday Lake           11.06923     de
## Hummingbird Lake       10.77328     de
## East Long Lake         10.26767      e
##
## attr(,"class")
## [1] "group"
```

16. From the findings above, which lakes have the same mean temperature, statistically speaking, as Peter Lake? Does any lake have a mean temperature that is statistically distinct from all the other lakes?

Answer: From the group test, we can tell that Ward Lake and Paul Lake have the same mean temperature, statistically speaking, as Peter Lake. We can tell that because these two lakes have the same letter as Peter Lake in “groups”.

17. If we were just looking at Peter Lake and Paul Lake. What’s another test we might explore to see whether they have distinct mean temperatures?

Answer: I did a little bit research online and found that another test we can conduct to explore the difference between mean temperature of Peter and Paul Lake is a paired t-test. It is commonly

used to determine whether there is a significant difference between the means of two related or “paired” groups. This method works better for comparing Peter Lake and Paul Lake since they are in the same “group”.

18. Wrangle the July data to include only records for Crampton Lake and Ward Lake. Run the two-sample T-test on these data to determine whether their July temperature are same or different. What does the test say? Are the mean temperatures for the lakes equal? Does that match you answer for part 16?

```
crampton.ward.july <- chem.phys.july %>%
  filter(lakename == "Crampton Lake" | lakename == "Ward Lake")
crampton.ward.july
```

| ##    | lakename      | year4 | daynum | depth | temperature_C |
|-------|---------------|-------|--------|-------|---------------|
| ## 1  | Crampton Lake | 1999  | 196    | 0.00  | 22.8          |
| ## 2  | Crampton Lake | 1999  | 196    | 0.50  | 22.6          |
| ## 3  | Crampton Lake | 1999  | 196    | 1.00  | 22.4          |
| ## 4  | Crampton Lake | 1999  | 196    | 1.50  | 22.2          |
| ## 5  | Crampton Lake | 1999  | 196    | 2.00  | 22.0          |
| ## 6  | Crampton Lake | 1999  | 196    | 2.50  | 21.9          |
| ## 7  | Crampton Lake | 1999  | 196    | 3.00  | 21.7          |
| ## 8  | Crampton Lake | 1999  | 196    | 3.50  | 21.3          |
| ## 9  | Crampton Lake | 1999  | 196    | 4.00  | 20.0          |
| ## 10 | Crampton Lake | 1999  | 196    | 4.50  | 17.1          |
| ## 11 | Crampton Lake | 1999  | 196    | 5.00  | 14.4          |
| ## 12 | Crampton Lake | 1999  | 196    | 5.50  | 12.1          |
| ## 13 | Crampton Lake | 1999  | 196    | 6.00  | 10.2          |
| ## 14 | Crampton Lake | 1999  | 196    | 6.50  | 9.1           |
| ## 15 | Crampton Lake | 1999  | 196    | 7.00  | 8.4           |
| ## 16 | Crampton Lake | 1999  | 196    | 8.00  | 7.1           |
| ## 17 | Crampton Lake | 1999  | 196    | 9.00  | 6.3           |
| ## 18 | Crampton Lake | 1999  | 196    | 10.00 | 5.8           |
| ## 19 | Crampton Lake | 1999  | 196    | 11.00 | 5.5           |
| ## 20 | Crampton Lake | 1999  | 196    | 12.00 | 5.4           |
| ## 21 | Crampton Lake | 2003  | 183    | 0.00  | 21.8          |
| ## 22 | Crampton Lake | 2003  | 183    | 0.50  | 21.9          |
| ## 23 | Crampton Lake | 2003  | 183    | 1.00  | 21.9          |
| ## 24 | Crampton Lake | 2003  | 183    | 1.50  | 21.8          |
| ## 25 | Crampton Lake | 2003  | 183    | 2.00  | 21.8          |
| ## 26 | Crampton Lake | 2003  | 183    | 2.50  | 21.3          |
| ## 27 | Crampton Lake | 2003  | 183    | 3.00  | 20.8          |
| ## 28 | Crampton Lake | 2003  | 183    | 3.50  | 19.9          |
| ## 29 | Crampton Lake | 2003  | 183    | 4.00  | 17.2          |
| ## 30 | Crampton Lake | 2003  | 183    | 4.50  | 15.2          |
| ## 31 | Crampton Lake | 2003  | 183    | 5.00  | 12.7          |
| ## 32 | Crampton Lake | 2003  | 183    | 5.50  | 11.2          |
| ## 33 | Crampton Lake | 2003  | 183    | 6.00  | 10.2          |
| ## 34 | Crampton Lake | 2003  | 183    | 6.50  | 8.8           |
| ## 35 | Crampton Lake | 2003  | 183    | 7.00  | 8.2           |
| ## 36 | Crampton Lake | 2003  | 183    | 8.00  | 6.8           |
| ## 37 | Crampton Lake | 2003  | 183    | 9.00  | 6.0           |
| ## 38 | Crampton Lake | 2003  | 183    | 10.00 | 5.4           |
| ## 39 | Crampton Lake | 2003  | 183    | 11.00 | 5.1           |

|       |               |      |     |       |      |
|-------|---------------|------|-----|-------|------|
| ## 40 | Crampton Lake | 2003 | 183 | 12.00 | 5.0  |
| ## 41 | Crampton Lake | 2003 | 197 | 0.00  | 22.2 |
| ## 42 | Crampton Lake | 2003 | 197 | 0.50  | 21.9 |
| ## 43 | Crampton Lake | 2003 | 197 | 1.00  | 21.8 |
| ## 44 | Crampton Lake | 2003 | 197 | 1.50  | 21.8 |
| ## 45 | Crampton Lake | 2003 | 197 | 2.00  | 21.7 |
| ## 46 | Crampton Lake | 2003 | 197 | 2.50  | 21.7 |
| ## 47 | Crampton Lake | 2003 | 197 | 3.00  | 21.7 |
| ## 48 | Crampton Lake | 2003 | 197 | 3.50  | 21.6 |
| ## 49 | Crampton Lake | 2003 | 197 | 4.00  | 19.6 |
| ## 50 | Crampton Lake | 2003 | 197 | 4.50  | 17.3 |
| ## 51 | Crampton Lake | 2003 | 197 | 5.00  | 14.5 |
| ## 52 | Crampton Lake | 2003 | 197 | 5.50  | 12.5 |
| ## 53 | Crampton Lake | 2003 | 197 | 6.00  | 10.9 |
| ## 54 | Crampton Lake | 2003 | 197 | 6.50  | 9.8  |
| ## 55 | Crampton Lake | 2003 | 197 | 7.00  | 8.7  |
| ## 56 | Crampton Lake | 2003 | 197 | 8.00  | 7.0  |
| ## 57 | Crampton Lake | 2003 | 197 | 9.00  | 6.0  |
| ## 58 | Crampton Lake | 2003 | 197 | 10.00 | 5.5  |
| ## 59 | Crampton Lake | 2003 | 197 | 11.00 | 5.2  |
| ## 60 | Crampton Lake | 2003 | 197 | 12.00 | 5.1  |
| ## 61 | Crampton Lake | 2003 | 211 | 0.00  | 23.2 |
| ## 62 | Crampton Lake | 2003 | 211 | 0.50  | 23.3 |
| ## 63 | Crampton Lake | 2003 | 211 | 1.00  | 23.2 |
| ## 64 | Crampton Lake | 2003 | 211 | 1.50  | 23.3 |
| ## 65 | Crampton Lake | 2003 | 211 | 2.00  | 23.3 |
| ## 66 | Crampton Lake | 2003 | 211 | 2.50  | 23.3 |
| ## 67 | Crampton Lake | 2003 | 211 | 3.00  | 23.3 |
| ## 68 | Crampton Lake | 2003 | 211 | 3.50  | 23.2 |
| ## 69 | Crampton Lake | 2003 | 211 | 4.00  | 22.4 |
| ## 70 | Crampton Lake | 2003 | 211 | 4.50  | 20.6 |
| ## 71 | Crampton Lake | 2003 | 211 | 5.00  | 17.3 |
| ## 72 | Crampton Lake | 2003 | 211 | 5.50  | 14.6 |
| ## 73 | Crampton Lake | 2003 | 211 | 6.00  | 12.1 |
| ## 74 | Crampton Lake | 2003 | 211 | 6.50  | 11.0 |
| ## 75 | Crampton Lake | 2003 | 211 | 7.00  | 9.6  |
| ## 76 | Crampton Lake | 2003 | 211 | 8.00  | 7.4  |
| ## 77 | Crampton Lake | 2003 | 211 | 9.00  | 6.4  |
| ## 78 | Crampton Lake | 2003 | 211 | 10.00 | 5.7  |
| ## 79 | Crampton Lake | 2003 | 211 | 11.00 | 5.4  |
| ## 80 | Crampton Lake | 2003 | 211 | 12.00 | 5.2  |
| ## 81 | Crampton Lake | 2004 | 183 | 0.00  | 19.8 |
| ## 82 | Crampton Lake | 2004 | 183 | 0.50  | 19.8 |
| ## 83 | Crampton Lake | 2004 | 183 | 1.00  | 19.8 |
| ## 84 | Crampton Lake | 2004 | 183 | 1.50  | 19.8 |
| ## 85 | Crampton Lake | 2004 | 183 | 2.00  | 19.7 |
| ## 86 | Crampton Lake | 2004 | 183 | 2.50  | 19.5 |
| ## 87 | Crampton Lake | 2004 | 183 | 3.00  | 18.9 |
| ## 88 | Crampton Lake | 2004 | 183 | 3.50  | 18.4 |
| ## 89 | Crampton Lake | 2004 | 183 | 4.00  | 18.0 |
| ## 90 | Crampton Lake | 2004 | 183 | 4.50  | 17.5 |
| ## 91 | Crampton Lake | 2004 | 183 | 5.00  | 15.6 |
| ## 92 | Crampton Lake | 2004 | 183 | 5.50  | 13.4 |
| ## 93 | Crampton Lake | 2004 | 183 | 6.00  | 11.2 |

|        |               |      |     |       |      |
|--------|---------------|------|-----|-------|------|
| ## 94  | Crampton Lake | 2004 | 183 | 6.50  | 10.2 |
| ## 95  | Crampton Lake | 2004 | 183 | 7.00  | 9.0  |
| ## 96  | Crampton Lake | 2004 | 183 | 8.00  | 7.6  |
| ## 97  | Crampton Lake | 2004 | 183 | 9.00  | 6.6  |
| ## 98  | Crampton Lake | 2004 | 183 | 10.00 | 6.0  |
| ## 99  | Crampton Lake | 2004 | 183 | 11.00 | 5.6  |
| ## 100 | Crampton Lake | 2004 | 183 | 12.00 | 5.5  |
| ## 101 | Crampton Lake | 2004 | 183 | 13.00 | 5.5  |
| ## 102 | Crampton Lake | 2004 | 183 | 14.00 | 5.5  |
| ## 103 | Crampton Lake | 2004 | 191 | 0.00  | 18.1 |
| ## 104 | Crampton Lake | 2004 | 191 | 0.50  | 18.1 |
| ## 105 | Crampton Lake | 2004 | 191 | 1.00  | 18.0 |
| ## 106 | Crampton Lake | 2004 | 191 | 1.50  | 18.0 |
| ## 107 | Crampton Lake | 2004 | 191 | 2.00  | 17.9 |
| ## 108 | Crampton Lake | 2004 | 191 | 2.50  | 17.9 |
| ## 109 | Crampton Lake | 2004 | 191 | 3.00  | 17.9 |
| ## 110 | Crampton Lake | 2004 | 191 | 3.50  | 17.8 |
| ## 111 | Crampton Lake | 2004 | 191 | 4.00  | 17.6 |
| ## 112 | Crampton Lake | 2004 | 191 | 4.50  | 17.4 |
| ## 113 | Crampton Lake | 2004 | 191 | 5.00  | 16.6 |
| ## 114 | Crampton Lake | 2004 | 191 | 5.50  | 13.8 |
| ## 115 | Crampton Lake | 2004 | 191 | 6.00  | 12.1 |
| ## 116 | Crampton Lake | 2004 | 191 | 6.50  | 10.9 |
| ## 117 | Crampton Lake | 2004 | 191 | 7.00  | 9.4  |
| ## 118 | Crampton Lake | 2004 | 191 | 8.00  | 7.6  |
| ## 119 | Crampton Lake | 2004 | 191 | 9.00  | 6.6  |
| ## 120 | Crampton Lake | 2004 | 191 | 10.00 | 6.0  |
| ## 121 | Crampton Lake | 2004 | 191 | 11.00 | 5.7  |
| ## 122 | Crampton Lake | 2004 | 191 | 12.00 | 5.6  |
| ## 123 | Crampton Lake | 2004 | 191 | 13.00 | 5.5  |
| ## 124 | Crampton Lake | 2004 | 191 | 14.00 | 5.5  |
| ## 125 | Crampton Lake | 2004 | 197 | 0.00  | 21.8 |
| ## 126 | Crampton Lake | 2004 | 197 | 0.50  | 21.8 |
| ## 127 | Crampton Lake | 2004 | 197 | 1.00  | 21.6 |
| ## 128 | Crampton Lake | 2004 | 197 | 1.50  | 21.6 |
| ## 129 | Crampton Lake | 2004 | 197 | 2.00  | 21.5 |
| ## 130 | Crampton Lake | 2004 | 197 | 2.50  | 20.9 |
| ## 131 | Crampton Lake | 2004 | 197 | 3.00  | 19.5 |
| ## 132 | Crampton Lake | 2004 | 197 | 3.50  | 19.1 |
| ## 133 | Crampton Lake | 2004 | 197 | 4.00  | 18.6 |
| ## 134 | Crampton Lake | 2004 | 197 | 4.50  | 17.9 |
| ## 135 | Crampton Lake | 2004 | 197 | 5.00  | 16.7 |
| ## 136 | Crampton Lake | 2004 | 197 | 5.50  | 14.8 |
| ## 137 | Crampton Lake | 2004 | 197 | 6.00  | 12.8 |
| ## 138 | Crampton Lake | 2004 | 197 | 6.50  | 11.1 |
| ## 139 | Crampton Lake | 2004 | 197 | 7.00  | 9.7  |
| ## 140 | Crampton Lake | 2004 | 197 | 8.00  | 7.7  |
| ## 141 | Crampton Lake | 2004 | 197 | 9.00  | 6.6  |
| ## 142 | Crampton Lake | 2004 | 197 | 10.00 | 6.1  |
| ## 143 | Crampton Lake | 2004 | 197 | 11.00 | 5.7  |
| ## 144 | Crampton Lake | 2004 | 197 | 12.00 | 5.6  |
| ## 145 | Crampton Lake | 2004 | 197 | 13.00 | 5.6  |
| ## 146 | Crampton Lake | 2004 | 197 | 14.00 | 5.5  |
| ## 147 | Crampton Lake | 2004 | 204 | 0.00  | 23.8 |

|        |               |      |     |       |      |
|--------|---------------|------|-----|-------|------|
| ## 148 | Crampton Lake | 2004 | 204 | 0.50  | 23.8 |
| ## 149 | Crampton Lake | 2004 | 204 | 1.00  | 23.8 |
| ## 150 | Crampton Lake | 2004 | 204 | 1.50  | 23.8 |
| ## 151 | Crampton Lake | 2004 | 204 | 2.00  | 23.8 |
| ## 152 | Crampton Lake | 2004 | 204 | 2.50  | 23.2 |
| ## 153 | Crampton Lake | 2004 | 204 | 3.00  | 22.0 |
| ## 154 | Crampton Lake | 2004 | 204 | 3.50  | 20.5 |
| ## 155 | Crampton Lake | 2004 | 204 | 4.00  | 19.9 |
| ## 156 | Crampton Lake | 2004 | 204 | 4.50  | 18.5 |
| ## 157 | Crampton Lake | 2004 | 204 | 5.00  | 17.3 |
| ## 158 | Crampton Lake | 2004 | 204 | 5.50  | 15.5 |
| ## 159 | Crampton Lake | 2004 | 204 | 6.00  | 13.5 |
| ## 160 | Crampton Lake | 2004 | 204 | 6.50  | 11.3 |
| ## 161 | Crampton Lake | 2004 | 204 | 7.00  | 10.1 |
| ## 162 | Crampton Lake | 2004 | 204 | 8.00  | 7.8  |
| ## 163 | Crampton Lake | 2004 | 204 | 9.00  | 6.8  |
| ## 164 | Crampton Lake | 2004 | 204 | 10.00 | 6.2  |
| ## 165 | Crampton Lake | 2004 | 204 | 11.00 | 5.8  |
| ## 166 | Crampton Lake | 2004 | 204 | 12.00 | 5.7  |
| ## 167 | Crampton Lake | 2004 | 204 | 13.00 | 5.6  |
| ## 168 | Crampton Lake | 2004 | 204 | 14.00 | 5.6  |
| ## 169 | Crampton Lake | 2004 | 211 | 0.00  | 22.7 |
| ## 170 | Crampton Lake | 2004 | 211 | 0.50  | 22.7 |
| ## 171 | Crampton Lake | 2004 | 211 | 1.00  | 22.7 |
| ## 172 | Crampton Lake | 2004 | 211 | 1.50  | 22.7 |
| ## 173 | Crampton Lake | 2004 | 211 | 2.00  | 22.7 |
| ## 174 | Crampton Lake | 2004 | 211 | 2.50  | 22.7 |
| ## 175 | Crampton Lake | 2004 | 211 | 3.00  | 22.7 |
| ## 176 | Crampton Lake | 2004 | 211 | 3.50  | 22.6 |
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| ## 182 | Crampton Lake | 2004 | 211 | 6.50  | 11.5 |
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| ## 184 | Crampton Lake | 2004 | 211 | 8.00  | 7.8  |
| ## 185 | Crampton Lake | 2004 | 211 | 9.00  | 6.8  |
| ## 186 | Crampton Lake | 2004 | 211 | 10.00 | 6.2  |
| ## 187 | Crampton Lake | 2004 | 211 | 11.00 | 5.8  |
| ## 188 | Crampton Lake | 2004 | 211 | 12.00 | 5.7  |
| ## 189 | Crampton Lake | 2004 | 211 | 13.00 | 5.6  |
| ## 190 | Crampton Lake | 2004 | 211 | 14.00 | 5.6  |
| ## 191 | Crampton Lake | 2005 | 186 | 0.00  | 22.3 |
| ## 192 | Crampton Lake | 2005 | 186 | 0.50  | 22.3 |
| ## 193 | Crampton Lake | 2005 | 186 | 1.00  | 22.4 |
| ## 194 | Crampton Lake | 2005 | 186 | 1.50  | 22.4 |
| ## 195 | Crampton Lake | 2005 | 186 | 2.00  | 22.4 |
| ## 196 | Crampton Lake | 2005 | 186 | 2.50  | 22.4 |
| ## 197 | Crampton Lake | 2005 | 186 | 3.00  | 22.4 |
| ## 198 | Crampton Lake | 2005 | 186 | 3.50  | 22.3 |
| ## 199 | Crampton Lake | 2005 | 186 | 4.00  | 22.3 |
| ## 200 | Crampton Lake | 2005 | 186 | 4.50  | 21.5 |
| ## 201 | Crampton Lake | 2005 | 186 | 5.00  | 17.9 |

|    |     |               |      |     |       |      |
|----|-----|---------------|------|-----|-------|------|
| ## | 202 | Crampton Lake | 2005 | 186 | 5.50  | 15.3 |
| ## | 203 | Crampton Lake | 2005 | 186 | 6.00  | 13.5 |
| ## | 204 | Crampton Lake | 2005 | 186 | 6.50  | 12.3 |
| ## | 205 | Crampton Lake | 2005 | 186 | 7.00  | 10.8 |
| ## | 206 | Crampton Lake | 2005 | 186 | 8.00  | 8.8  |
| ## | 207 | Crampton Lake | 2005 | 186 | 9.00  | 7.4  |
| ## | 208 | Crampton Lake | 2005 | 186 | 10.00 | 6.0  |
| ## | 209 | Crampton Lake | 2005 | 186 | 11.00 | 5.8  |
| ## | 210 | Crampton Lake | 2005 | 186 | 12.00 | 5.7  |
| ## | 211 | Crampton Lake | 2005 | 192 | 0.00  | 25.7 |
| ## | 212 | Crampton Lake | 2005 | 192 | 0.50  | 25.7 |
| ## | 213 | Crampton Lake | 2005 | 192 | 1.00  | 25.4 |
| ## | 214 | Crampton Lake | 2005 | 192 | 1.50  | 25.4 |
| ## | 215 | Crampton Lake | 2005 | 192 | 2.00  | 25.3 |
| ## | 216 | Crampton Lake | 2005 | 192 | 2.50  | 24.2 |
| ## | 217 | Crampton Lake | 2005 | 192 | 3.00  | 23.5 |
| ## | 218 | Crampton Lake | 2005 | 192 | 3.50  | 22.9 |
| ## | 219 | Crampton Lake | 2005 | 192 | 4.00  | 22.5 |
| ## | 220 | Crampton Lake | 2005 | 192 | 4.50  | 21.4 |
| ## | 221 | Crampton Lake | 2005 | 192 | 5.00  | 19.5 |
| ## | 222 | Crampton Lake | 2005 | 192 | 5.50  | 16.5 |
| ## | 223 | Crampton Lake | 2005 | 192 | 6.00  | 14.6 |
| ## | 224 | Crampton Lake | 2005 | 192 | 6.50  | 12.9 |
| ## | 225 | Crampton Lake | 2005 | 192 | 7.00  | 11.4 |
| ## | 226 | Crampton Lake | 2005 | 192 | 8.00  | 9.0  |
| ## | 227 | Crampton Lake | 2005 | 192 | 9.00  | 7.5  |
| ## | 228 | Crampton Lake | 2005 | 192 | 10.00 | 6.5  |
| ## | 229 | Crampton Lake | 2005 | 192 | 11.00 | 6.0  |
| ## | 230 | Crampton Lake | 2005 | 192 | 12.00 | 5.7  |
| ## | 231 | Crampton Lake | 2005 | 192 | 13.00 | 5.6  |
| ## | 232 | Crampton Lake | 2005 | 199 | 0.00  | 27.5 |
| ## | 233 | Crampton Lake | 2005 | 199 | 0.50  | 27.5 |
| ## | 234 | Crampton Lake | 2005 | 199 | 1.00  | 27.5 |
| ## | 235 | Crampton Lake | 2005 | 199 | 1.50  | 27.4 |
| ## | 236 | Crampton Lake | 2005 | 199 | 2.00  | 27.4 |
| ## | 237 | Crampton Lake | 2005 | 199 | 2.50  | 27.4 |
| ## | 238 | Crampton Lake | 2005 | 199 | 3.00  | 26.0 |
| ## | 239 | Crampton Lake | 2005 | 199 | 3.50  | 24.8 |
| ## | 240 | Crampton Lake | 2005 | 199 | 4.00  | 23.7 |
| ## | 241 | Crampton Lake | 2005 | 199 | 4.50  | 22.5 |
| ## | 242 | Crampton Lake | 2005 | 199 | 5.00  | 19.8 |
| ## | 243 | Crampton Lake | 2005 | 199 | 5.50  | 17.1 |
| ## | 244 | Crampton Lake | 2005 | 199 | 6.00  | 15.3 |
| ## | 245 | Crampton Lake | 2005 | 199 | 6.50  | 13.5 |
| ## | 246 | Crampton Lake | 2005 | 199 | 7.00  | 12.0 |
| ## | 247 | Crampton Lake | 2005 | 199 | 8.00  | 9.0  |
| ## | 248 | Crampton Lake | 2005 | 199 | 9.00  | 7.7  |
| ## | 249 | Crampton Lake | 2005 | 199 | 10.00 | 6.6  |
| ## | 250 | Crampton Lake | 2005 | 199 | 11.00 | 6.1  |
| ## | 251 | Crampton Lake | 2005 | 199 | 12.00 | 5.7  |
| ## | 252 | Crampton Lake | 2005 | 199 | 13.00 | 5.7  |
| ## | 253 | Crampton Lake | 2005 | 206 | 0.00  | 25.3 |
| ## | 254 | Crampton Lake | 2005 | 206 | 0.50  | 25.1 |
| ## | 255 | Crampton Lake | 2005 | 206 | 1.00  | 25.0 |

|    |     |               |      |     |       |      |
|----|-----|---------------|------|-----|-------|------|
| ## | 256 | Crampton Lake | 2005 | 206 | 1.50  | 24.9 |
| ## | 257 | Crampton Lake | 2005 | 206 | 2.00  | 24.9 |
| ## | 258 | Crampton Lake | 2005 | 206 | 2.50  | 24.9 |
| ## | 259 | Crampton Lake | 2005 | 206 | 3.00  | 24.8 |
| ## | 260 | Crampton Lake | 2005 | 206 | 3.50  | 24.8 |
| ## | 261 | Crampton Lake | 2005 | 206 | 4.00  | 24.7 |
| ## | 262 | Crampton Lake | 2005 | 206 | 4.50  | 24.5 |
| ## | 263 | Crampton Lake | 2005 | 206 | 5.00  | 21.3 |
| ## | 264 | Crampton Lake | 2005 | 206 | 5.50  | 18.8 |
| ## | 265 | Crampton Lake | 2005 | 206 | 6.00  | 16.7 |
| ## | 266 | Crampton Lake | 2005 | 206 | 6.50  | 14.4 |
| ## | 267 | Crampton Lake | 2005 | 206 | 7.00  | 12.3 |
| ## | 268 | Crampton Lake | 2005 | 206 | 8.00  | 9.9  |
| ## | 269 | Crampton Lake | 2005 | 206 | 9.00  | 7.9  |
| ## | 270 | Crampton Lake | 2005 | 206 | 10.00 | 6.8  |
| ## | 271 | Crampton Lake | 2005 | 206 | 11.00 | 6.1  |
| ## | 272 | Crampton Lake | 2005 | 206 | 12.00 | 5.8  |
| ## | 273 | Crampton Lake | 2005 | 206 | 13.00 | 5.7  |
| ## | 274 | Crampton Lake | 2006 | 186 | 0.00  | 22.7 |
| ## | 275 | Crampton Lake | 2006 | 186 | 0.50  | 22.8 |
| ## | 276 | Crampton Lake | 2006 | 186 | 1.00  | 22.8 |
| ## | 277 | Crampton Lake | 2006 | 186 | 1.50  | 22.8 |
| ## | 278 | Crampton Lake | 2006 | 186 | 2.00  | 22.8 |
| ## | 279 | Crampton Lake | 2006 | 186 | 2.50  | 22.5 |
| ## | 280 | Crampton Lake | 2006 | 186 | 3.00  | 22.6 |
| ## | 281 | Crampton Lake | 2006 | 186 | 3.50  | 22.6 |
| ## | 282 | Crampton Lake | 2006 | 186 | 4.00  | 20.3 |
| ## | 283 | Crampton Lake | 2006 | 186 | 4.50  | 17.4 |
| ## | 284 | Crampton Lake | 2006 | 186 | 5.00  | 15.2 |
| ## | 285 | Crampton Lake | 2006 | 186 | 5.50  | 13.7 |
| ## | 286 | Crampton Lake | 2006 | 186 | 6.00  | 12.1 |
| ## | 287 | Crampton Lake | 2006 | 186 | 6.50  | 10.8 |
| ## | 288 | Crampton Lake | 2006 | 186 | 7.00  | 9.6  |
| ## | 289 | Crampton Lake | 2006 | 186 | 7.50  | 8.3  |
| ## | 290 | Crampton Lake | 2006 | 186 | 8.00  | 7.4  |
| ## | 291 | Crampton Lake | 2006 | 186 | 8.50  | 6.3  |
| ## | 292 | Crampton Lake | 2006 | 186 | 9.00  | 6.0  |
| ## | 293 | Crampton Lake | 2006 | 186 | 10.00 | 5.5  |
| ## | 294 | Crampton Lake | 2006 | 186 | 11.00 | 5.2  |
| ## | 295 | Crampton Lake | 2006 | 186 | 12.00 | 5.1  |
| ## | 296 | Crampton Lake | 2006 | 200 | 0.00  | 25.4 |
| ## | 297 | Crampton Lake | 2006 | 200 | 0.50  | 25.4 |
| ## | 298 | Crampton Lake | 2006 | 200 | 1.00  | 25.4 |
| ## | 299 | Crampton Lake | 2006 | 200 | 1.50  | 25.4 |
| ## | 300 | Crampton Lake | 2006 | 200 | 2.00  | 25.4 |
| ## | 301 | Crampton Lake | 2006 | 200 | 2.50  | 25.4 |
| ## | 302 | Crampton Lake | 2006 | 200 | 3.00  | 25.4 |
| ## | 303 | Crampton Lake | 2006 | 200 | 3.50  | 24.1 |
| ## | 304 | Crampton Lake | 2006 | 200 | 4.00  | 22.5 |
| ## | 305 | Crampton Lake | 2006 | 200 | 4.50  | 19.8 |
| ## | 306 | Crampton Lake | 2006 | 200 | 5.00  | 17.5 |
| ## | 307 | Crampton Lake | 2006 | 200 | 5.50  | 15.0 |
| ## | 308 | Crampton Lake | 2006 | 200 | 6.00  | 13.3 |
| ## | 309 | Crampton Lake | 2006 | 200 | 6.50  | 11.4 |

|    |     |               |      |     |       |      |
|----|-----|---------------|------|-----|-------|------|
| ## | 310 | Crampton Lake | 2006 | 200 | 7.00  | 9.6  |
| ## | 311 | Crampton Lake | 2006 | 200 | 7.50  | 8.7  |
| ## | 312 | Crampton Lake | 2006 | 200 | 8.00  | 7.5  |
| ## | 313 | Crampton Lake | 2006 | 200 | 8.50  | 6.7  |
| ## | 314 | Crampton Lake | 2006 | 200 | 9.00  | 6.4  |
| ## | 315 | Crampton Lake | 2006 | 200 | 10.00 | 5.7  |
| ## | 316 | Crampton Lake | 2006 | 200 | 11.00 | 5.4  |
| ## | 317 | Crampton Lake | 2006 | 200 | 12.00 | 5.1  |
| ## | 318 | Crampton Lake | 2006 | 200 | 13.00 | 5.1  |
| ## | 319 | Ward Lake     | 2010 | 188 | 0.00  | 25.5 |
| ## | 320 | Ward Lake     | 2010 | 188 | 0.50  | 25.3 |
| ## | 321 | Ward Lake     | 2010 | 188 | 0.75  | 24.9 |
| ## | 322 | Ward Lake     | 2010 | 188 | 1.00  | 24.8 |
| ## | 323 | Ward Lake     | 2010 | 188 | 1.50  | 23.7 |
| ## | 324 | Ward Lake     | 2010 | 188 | 2.00  | 21.8 |
| ## | 325 | Ward Lake     | 2010 | 188 | 2.50  | 18.6 |
| ## | 326 | Ward Lake     | 2010 | 188 | 3.00  | 16.6 |
| ## | 327 | Ward Lake     | 2010 | 188 | 3.50  | 14.6 |
| ## | 328 | Ward Lake     | 2010 | 188 | 4.00  | 12.5 |
| ## | 329 | Ward Lake     | 2010 | 188 | 4.50  | 10.4 |
| ## | 330 | Ward Lake     | 2010 | 188 | 5.00  | 9.0  |
| ## | 331 | Ward Lake     | 2010 | 188 | 5.50  | 7.7  |
| ## | 332 | Ward Lake     | 2010 | 188 | 6.00  | 7.1  |
| ## | 333 | Ward Lake     | 2010 | 188 | 6.50  | 6.9  |
| ## | 334 | Ward Lake     | 2010 | 188 | 6.75  | 6.7  |
| ## | 335 | Ward Lake     | 2010 | 195 | 0.00  | 24.1 |
| ## | 336 | Ward Lake     | 2010 | 195 | 0.50  | 24.1 |
| ## | 337 | Ward Lake     | 2010 | 195 | 1.00  | 24.0 |
| ## | 338 | Ward Lake     | 2010 | 195 | 1.50  | 23.9 |
| ## | 339 | Ward Lake     | 2010 | 195 | 2.00  | 23.2 |
| ## | 340 | Ward Lake     | 2010 | 195 | 2.50  | 20.3 |
| ## | 341 | Ward Lake     | 2010 | 195 | 3.00  | 17.4 |
| ## | 342 | Ward Lake     | 2010 | 195 | 3.50  | 14.7 |
| ## | 343 | Ward Lake     | 2010 | 195 | 4.00  | 12.5 |
| ## | 344 | Ward Lake     | 2010 | 195 | 4.50  | 10.6 |
| ## | 345 | Ward Lake     | 2010 | 195 | 5.00  | 8.8  |
| ## | 346 | Ward Lake     | 2010 | 195 | 5.50  | 7.7  |
| ## | 347 | Ward Lake     | 2010 | 195 | 6.00  | 7.2  |
| ## | 348 | Ward Lake     | 2010 | 195 | 6.50  | 7.0  |
| ## | 349 | Ward Lake     | 2010 | 202 | 0.00  | 23.9 |
| ## | 350 | Ward Lake     | 2010 | 202 | 0.50  | 23.8 |
| ## | 351 | Ward Lake     | 2010 | 202 | 1.00  | 23.7 |
| ## | 352 | Ward Lake     | 2010 | 202 | 1.50  | 23.3 |
| ## | 353 | Ward Lake     | 2010 | 202 | 2.00  | 23.1 |
| ## | 354 | Ward Lake     | 2010 | 202 | 2.50  | 20.7 |
| ## | 355 | Ward Lake     | 2010 | 202 | 3.00  | 17.0 |
| ## | 356 | Ward Lake     | 2010 | 202 | 3.50  | 15.1 |
| ## | 357 | Ward Lake     | 2010 | 202 | 4.00  | 12.6 |
| ## | 358 | Ward Lake     | 2010 | 202 | 4.50  | 10.7 |
| ## | 359 | Ward Lake     | 2010 | 202 | 5.00  | 9.1  |
| ## | 360 | Ward Lake     | 2010 | 202 | 5.50  | 7.9  |
| ## | 361 | Ward Lake     | 2010 | 202 | 6.00  | 7.3  |
| ## | 362 | Ward Lake     | 2010 | 202 | 6.50  | 7.1  |
| ## | 363 | Ward Lake     | 2010 | 202 | 7.00  | 7.1  |



|        |           |      |     |      |      |
|--------|-----------|------|-----|------|------|
| ## 364 | Ward Lake | 2010 | 209 | 0.00 | 23.6 |
| ## 365 | Ward Lake | 2010 | 209 | 0.50 | 23.6 |
| ## 366 | Ward Lake | 2010 | 209 | 1.00 | 23.5 |
| ## 367 | Ward Lake | 2010 | 209 | 1.50 | 23.4 |
| ## 368 | Ward Lake | 2010 | 209 | 2.00 | 23.1 |
| ## 369 | Ward Lake | 2010 | 209 | 2.50 | 21.0 |
| ## 370 | Ward Lake | 2010 | 209 | 3.00 | 18.2 |
| ## 371 | Ward Lake | 2010 | 209 | 3.50 | 15.7 |
| ## 372 | Ward Lake | 2010 | 209 | 4.00 | 13.4 |
| ## 373 | Ward Lake | 2010 | 209 | 4.50 | 11.1 |
| ## 374 | Ward Lake | 2010 | 209 | 5.00 | 9.2  |
| ## 375 | Ward Lake | 2010 | 209 | 5.50 | 8.0  |
| ## 376 | Ward Lake | 2010 | 209 | 6.00 | 7.4  |
| ## 377 | Ward Lake | 2010 | 209 | 6.50 | 7.2  |
| ## 378 | Ward Lake | 2010 | 209 | 6.75 | 7.0  |
| ## 379 | Ward Lake | 2012 | 187 | 0.00 | 27.6 |
| ## 380 | Ward Lake | 2012 | 187 | 0.50 | 27.0 |
| ## 381 | Ward Lake | 2012 | 187 | 1.00 | 24.6 |
| ## 382 | Ward Lake | 2012 | 187 | 1.50 | 20.4 |
| ## 383 | Ward Lake | 2012 | 187 | 2.00 | 15.8 |
| ## 384 | Ward Lake | 2012 | 187 | 2.50 | 12.3 |
| ## 385 | Ward Lake | 2012 | 187 | 3.00 | 10.2 |
| ## 386 | Ward Lake | 2012 | 187 | 3.50 | 7.9  |
| ## 387 | Ward Lake | 2012 | 187 | 4.00 | 6.9  |
| ## 388 | Ward Lake | 2012 | 187 | 4.50 | 6.4  |
| ## 389 | Ward Lake | 2012 | 187 | 5.00 | 6.1  |
| ## 390 | Ward Lake | 2012 | 187 | 5.50 | 5.8  |
| ## 391 | Ward Lake | 2012 | 187 | 6.00 | 5.7  |
| ## 392 | Ward Lake | 2012 | 194 | 0.00 | 25.9 |
| ## 393 | Ward Lake | 2012 | 194 | 0.50 | 25.1 |
| ## 394 | Ward Lake | 2012 | 194 | 1.00 | 24.5 |
| ## 395 | Ward Lake | 2012 | 194 | 1.50 | 20.8 |
| ## 396 | Ward Lake | 2012 | 194 | 2.00 | 16.5 |
| ## 397 | Ward Lake | 2012 | 194 | 2.50 | 12.6 |
| ## 398 | Ward Lake | 2012 | 194 | 3.00 | 10.3 |
| ## 399 | Ward Lake | 2012 | 194 | 3.50 | 8.2  |
| ## 400 | Ward Lake | 2012 | 194 | 4.00 | 7.2  |
| ## 401 | Ward Lake | 2012 | 194 | 4.50 | 6.6  |
| ## 402 | Ward Lake | 2012 | 194 | 5.00 | 6.3  |
| ## 403 | Ward Lake | 2012 | 194 | 5.50 | 6.0  |
| ## 404 | Ward Lake | 2012 | 194 | 6.00 | 5.9  |
| ## 405 | Ward Lake | 2012 | 201 | 0.00 | 23.6 |
| ## 406 | Ward Lake | 2012 | 201 | 0.50 | 23.3 |
| ## 407 | Ward Lake | 2012 | 201 | 1.00 | 23.2 |
| ## 408 | Ward Lake | 2012 | 201 | 1.50 | 21.5 |
| ## 409 | Ward Lake | 2012 | 201 | 2.00 | 16.9 |
| ## 410 | Ward Lake | 2012 | 201 | 2.50 | 13.2 |
| ## 411 | Ward Lake | 2012 | 201 | 3.00 | 10.3 |
| ## 412 | Ward Lake | 2012 | 201 | 3.50 | 8.6  |
| ## 413 | Ward Lake | 2012 | 201 | 4.00 | 7.7  |
| ## 414 | Ward Lake | 2012 | 201 | 4.50 | 6.8  |
| ## 415 | Ward Lake | 2012 | 201 | 5.00 | 6.4  |
| ## 416 | Ward Lake | 2012 | 201 | 5.50 | 6.2  |
| ## 417 | Ward Lake | 2012 | 201 | 6.00 | 6.0  |

```
## 418    Ward Lake  2012    201  6.50          5.9
## 419    Ward Lake  2012    201  7.00          5.8
## 420    Ward Lake  2012    208  0.00         24.3
## 421    Ward Lake  2012    208  0.50         24.2
## 422    Ward Lake  2012    208  1.00         24.2
## 423    Ward Lake  2012    208  1.50         22.3
## 424    Ward Lake  2012    208  2.00         17.5
## 425    Ward Lake  2012    208  2.50         13.7
## 426    Ward Lake  2012    208  3.00         10.5
## 427    Ward Lake  2012    208  3.50          8.5
## 428    Ward Lake  2012    208  4.00          7.6
## 429    Ward Lake  2012    208  4.50          6.8
## 430    Ward Lake  2012    208  5.00          6.4
## 431    Ward Lake  2012    208  5.50          6.1
## 432    Ward Lake  2012    208  6.00          6.0
## 433    Ward Lake  2012    208  6.50          5.9
## 434    Ward Lake  2012    208  7.00          5.8
```

```
july.twosample <- t.test(crampton.ward.july$temperature_C ~ crampton.ward.july$lakename)
july.twosample
```

```
##
##  Welch Two Sample t-test
##
## data:  crampton.ward.july$temperature_C by crampton.ward.july$lakename
## t = 1.1181, df = 200.37, p-value = 0.2649
## alternative hypothesis: true difference in means between group Crampton Lake and group Ward Lake is not equal to 0
## 95 percent confidence interval:
##  -0.6821129  2.4686451
## sample estimates:
## mean in group Crampton Lake      mean in group Ward Lake
##           15.35189              14.45862
```

Answer: The results indicate that these two lakes have different mean temperature in July. Crampton has 15.35, and Ward has 14.46.